RetailMart Customer Intelligence Platform

Unit 5: Algorithm Comparison & Customer Segmentation

Data Science Team

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1 Algorithm Comparison: Churn Prediction

To answer the business question *"Can we predict which customers will stop shopping with us?"*, we benchmarked four classification algorithms using PySpark MLlib.

1.1 Experimental Setup

The models were trained on a leak-free 80/20 customer-level split of the cleaned dataset ($n_{train} = 778,429$; $n_{test} = 194,837$).

- **Target Variable:** Churn (22.3% positive class, 77.7% negative class).
- **Class Imbalance:** Addressed using a positive class weight of 3.48 to prevent the models from biasing toward the majority class.
- **Features used:** `total_spend`, `quantity`, `loyalty_years`, `num_visits`, `age`, `days_since_last`, `gender`, `payment_method`.
- **Excluded Features:** `order_count`, `avg_order_value`, and `distinct_categories` were removed due to zero variance (the dataset generator assigns a fresh `customer_id` per transaction, resulting in exactly one order per customer).

Runtime Note: To keep Colab CPU runtime tractable, tree ensembles were trained on a 35% subsample of the training set (272,450 rows). Evaluation was performed on the full 194,837-row test set to ensure fair comparison.

1.2 Model Results

Table 1 summarizes the performance of the four algorithms.

Table 1: Classifier Performance Comparison

Model	AUC-ROC	Accuracy	Precision	Recall	F1-Score	Train T
Gradient-Boosted Trees (GBT)	**0.713**	0.702	0.760	0.702	0.721	1
Random Forest	0.712	0.706	0.760	0.706	**0.725**	2
Logistic Regression	0.655	0.557	0.734	0.557	0.593	
Decision Tree	0.500	0.702	0.760	0.702	0.722	

1.3 Confusion Matrix Analysis

Evaluating the models at a 0.5 threshold on the test set provided further operational insight:

- **GBT** successfully identified 26,581 churners (True Positives) while generating 41,157 False Positives. This represents a 61% recall and 39% precision, offering the best balance for cost-sensitive marketing.
- **Logistic Regression** achieved the highest recall (71%) but generated an unacceptable 73,621 False Positives, which would result in significant budget waste for win-back campaigns.

1.4 The Decision Tree Failure Mode

The Decision Tree achieved 70% accuracy but collapsed to an AUC of 0.50. Reading the confusion matrix reveals this is a **ranking failure, not a classification failure**. The tree produces coarse, tied leaf probabilities on this weak-signal dataset. Because its probabilities cannot monotonically rank the minority class, the ROC curve degenerates to a staircase near the diagonal (random guess). Consequently, single trees are disqualified for scoring or prioritising customers.

Recommendation for Churn Prediction

Deploy **Gradient-Boosted Trees (GBT)**. It achieves the highest AUC (0.713) and an optimal precision/recall trade-off, running in less than half the training time of a Random Forest (113s vs 257s). Logistic Regression remains a viable fast-retraining baseline, while Decision Trees are disqualified for scoring.

2 Customer Segmentation

To answer *"Which customer segments should we target with marketing campaigns?"*, we applied unsupervised clustering to the standardized customer feature vectors.

2.1 K-Means vs. Bisecting K-Means

We evaluated clustering performance across $k = 2$ to $k = 7$ using the silhouette score.

- **K-Means** optimal performance was found at $k = 6$, yielding a silhouette score of **0.2834**.
- **Bisecting K-Means** achieved a significantly lower silhouette score of **0.1678**.

Therefore, standard K-Means ($k = 6$) was selected for the final segmentation.

2.2 Segment Profiles

Table 2 outlines the six discovered segments. Notably, five of the six clusters represent the "average" customer (churn rate $\approx 22\%$, ≈ 8 visits, ≈ 1.8 loyalty years).

Cluster 2 (20,591 customers) is the critical outlier segment. These are highly loyal customers with 22.7 visits and 8.2 loyalty years, resulting in a significantly lower churn rate of 14.4%.

Table 2: K-Means Segment Profiles ($k = 6$)

Cluster	n	Avg Spend	Avg Visits	Avg Loyalty (yrs)	Avg Age	Churn Rate
0	34,478	\$170.45	8.50	1.82	41.8	22.1%
1	22,955	\$171.68	8.46	1.81	41.9	24.6%
2	**20,591**	**\$171.92**	**22.77**	**8.23**	41.9	**14.4%**
3	47,591	\$172.53	8.04	1.60	41.7	25.0%
4	34,554	\$172.51	8.45	1.81	41.9	22.3%
5	34,692	\$171.02	8.53	1.83	41.9	22.2%

Marketing Targeting Strategy

- **Protect Cluster 2:** These are VIP customers. Target them with loyalty tier upgrades and early-access rewards to maintain their low churn rate.
- **Win-Back Clusters 1 & 3:** These groups exhibit the highest churn rates (24.6% and 25.0%) and lowest loyalty. Allocate win-back discount campaigns and re-engagement emails specifically to these segments.